Chathil Rajamanthree Electrical Engineering Student

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TECHNICAL SKILLS

Languages — SystemVerilog, VHDL, C, ARMv7/Assembly, Python, Bash/Shell scripting

Tools and EDA softwares — Cadence Virtuoso, Spectre, LTspice, Quartus, ModelSim, Git, Altium, Linux

Laboratory — FPGA, MCU, Soldering, Function generator, Multimeter, Oscilloscope, Logic Analyzer

EDUCATION

Bachelor of Applied Science - Electrical Engineering (Co-op), Sep 2023 – May 2028 | Vancouver, BC
University of British Columbia, CGPA: 80.8%
Relevant courses: Analog CMOS Integrated Circuit Design, Digital Systems Design, Signals and Systems

TECHNICAL WORK EXPERIENCE

Dialog Network Services, *Radio Network Planning Intern* Jul 2025 – Aug 2025 | Colombo, Sri Lanka

- Assisted deployment and reconfiguration of 2G/4G/5G base stations across urban and rural areas, with a focus on base station antenna interference mitigation to optimize network throughput to the end user
- Conducted drive tests with TEMS Investigation to assess antenna parameter effects on outdoor signal performance
- Performed indoor walk tests with G-NetTrack Pro to diagnose antenna complaints in existing buildings and verify signal coverage and quality in new building sites
- Proposed antenna upgrades for high-density public events, collaborating with external vendors to align solutions with technical antenna requirements

University of British Columbia, *Undergraduate Teaching Assistant* Jan 2025 – May 2025 | Vancouver, BC

- Offered individualized support to students in C programming and Arduino-based microcontroller development
- Debugged student code and piloted exam questions for APSC 160: Intro to Computation in Engineering Design

UBC Bionics, *Electrical Team Lead* Sep 2023 – Present | Vancouver, BC

- Leading a 12-member engineering subteam within a multidisciplinary student design team to develop GRASP, a bionic arm, focusing on EMG signal acquisition, battery management, and haptic boards with firmware.
- Developing a custom EMG acquisition board, progressing from simulation to PCB layout with emphasis on error debugging, power coordination, and optimized routing for low-noise operation.

PROJECTS

Two-stage 45nm CMOS Op-Amp Design with CMFB, *UBC ELEC 401* Nov 2025 – Dec 2025

- Designed a two-stage **differential to single-ended operational amplifier** in **GPDK45** using **Cadence EDA tools** to meet design requirements, achieving 52dB gain, 618MHz unity gain frequency, 65° phase margin, 0.65V output swing, and 48.45V/μs slew rate under 375μW of power consumption
- Applied **Miller compensation** with series RC zero placement to meet stability and bandwidth requirements.
- Implemented a **common-mode feedback loop** to regulate output common-mode voltage
- Performed **AC**, **transient** and **stability** analyses (Bode plots, step response, settling time and slew-rate characterization)

FPGA Digital Signal Processing, *UBC CPEN 311* Jun 2025

- Designed a digital communication system to enable hardware/software co-design in **SystemVerilog** and **embedded C** with the Nios II processor on the DE1-SoC to generate and view real-time ASK, BPSK and FSK modulation
- Integrated a Direct Digital Synthesis carrier signal generator and a 5-bit LFSR to modulate carrier waves with proper **clock domain crossing** techniques
- Synthesized FSK modulation using **embedded C** on the Nios II processor by generating interrupts using **Qsys** PIOs

FPGA Multi-Core RC4 Cracking Circuit, *UBC CPEN 311* Jun 2025

- Designed a **hardware-accelerated** brute-force attack on the RC4 stream cypher using 10 **parallel decryption cores** to achieve a 10x speedup over single-core cycling across a 24-bit keyspace
- Coordinated communication between 41 **finite state machines (FSMs)** using a standardised start-finish protocol with **SystemVerilog** and **VHDL**, validated using **SignalTap**

Proxmox Homelab Server, *Personal Project* May 2025 – Jul 2025

- Built a Proxmox-based virtualization environment hosting multiple **Linux** containers and virtual machines
- Automated game server deployment using custom **Bash scripts** with streamlined server management
- Deployed a Samba file server with a Tailscale VPN for secure remote file sharing across devices

Simple FPGA iPod, *UBC CPEN 311* May 2025

- Implemented multiple **glitchless FSMs** and a configurable clock divider to read from flash memory and control audio playback including real-time speed adjustments via a PS/2 keyboard
- Designed a 2-stage **synchronizer** to safely transfer signals across asynchronous clock domains
- Performed real-time averaging of audio samples using assembly and an **interrupt service routine (ISR)** on an embedded processor (**PicoBlaze**) to create an audio strength meter